

Integrating artificial-intelligence assistance into chemical process control education

Zuyi Huang 

Department of Chemical & Biological Engineering, Villanova University, Villanova, PA 19805, USA

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ABSTRACT

Chemical process control is traditionally one of the most mathematically demanding courses in the chemical engineering curriculum. Students need to master ordinary differential equations (ODEs), Laplace transforms, transfer functions and proportional-integral-derivative (PID) controller design. At Villanova University we developed a pilot project to integrate artificial-intelligence assistance throughout the process-control pipeline. Three modules were created: (1) Python coding for ODE simulation, (2) Laplace and inverse Laplace transforms with transfer-function parameter estimation, and (3) PID controller tuning. For each module, pre-class videos and handouts introduced theoretical concepts and demonstrated example code. Students reviewed these materials before class, and class time was used primarily for answering questions and clarifying concepts. Most project work was completed independently outside of class. Students were instructed to ask a large-language-model chatbot (ChatGPT) to explain equations, generate and debug Python code, and interpret the physical meaning of models. Pre-class handouts and videos prepared students for both in-class and independent practice, and anonymous surveys measured learning impact and students' overall experience. Fifty-nine students completed all three modules and responded to a 15-question survey, which included 13 multiple-choice rating questions and 2 open-ended questions. The rating questions were grouped into categories measuring students' confidence in Python coding, understanding of ODE modeling and process control concepts, ability to use AI tools effectively, and perceived applicability of the skills to real-world problems. Responses were scored on a five-point scale, where a rating of 5 indicated the most positive feedback. Statistical analysis showed mean ratings ranging from 3.28 to 4.36, with the highest confidence gain reported in the category assessing the ability to apply the learned skills to real-world problems. Qualitative comments reveal that students valued the novelty of using AI tools but desired clearer instructions and more coding guidance. This study demonstrates a fully integrated approach to AI-assisted education in chemical process control, spanning ODE modeling, Laplace transform, transfer functions, and controller tuning within a single coherent framework. By combining Python-based coding in Google Colab with structured AI interaction strategies, the work bridges theoretical concepts and practical applications across the entire modeling-control pipeline. This project offers an example for AI-based control and engineering education studies.

1. Introduction

Chemical process control seeks to maintain key process variables such as temperature, pressure and composition at desired setpoints despite disturbances. The most widely used feedback strategy is the proportional-integral-derivative (PID) controller. In a PID controller the control action is proportional to the current error between the setpoint and the measured variable, integrates past errors and anticipates future changes by differentiating the error (Bequette, 2006; Coughanowr and LeBlanc, 2009; Doyle, 1998; Seborg et al., 2016). Classical

texts and surveys emphasize that PID controllers remain the workhorse of process control because they are simple to implement and do not require explicit process models, although careful tuning is necessary to achieve good performance. Advanced techniques such as model predictive control and optimization-based strategies have become commonplace in industry, but they demand more sophisticated mathematical tools and higher computational cost (Wayne Bequette, 2018). The challenge for educators is to teach these topics in a way that balances theory with practice and to equip students with modern programming skills.

E-mail address: zuyi.huang@villanova.edu.

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While classical process control courses typically follow a lecture-centric format supported by textbooks and commercial software, chemical process control education has shifted over the past decade from traditional lecture-based delivery to active-learning pedagogies, with the flipped classroom emerging as a leading model. In this approach, students first review key concepts outside class, typically through videos, assigned readings, or online modules, and then use classroom time for interactive problem-solving and discussion. Bequette's work in process systems engineering is a well-documented example, combining a flipped format with a studio classroom environment to engage students in complex, realistic control problems (Wayne Bequette, 2018). In his courses, students finish readings and screencasts before class, take short online quizzes to ensure preparation, and then work in teams on numerical problem-solving tasks. This teaching structure has been shown to improve preparedness, inspire collaborative learning, and allow deeper engagement with challenging material (Koretsky, 2015; Wayne Bequette, 2018). Similar flipped and studio-based designs have been adopted in other programs, including fully online process control courses (Lewin and Barzilai, 2022) and hybrid models that pair pre-class videos with in-class exercises. For example, Villanova University applied an inverted-classroom format to reinforce computational skills for process control by combining pre-class MATLAB modules on ODE simulation, Laplace transforms, and controller design with in-class problem-solving activities, which improved both engagement and coding proficiency (Li and Huang, 2017).

These active-learning approaches are often complemented by simulation-based exercises and hands-on experiences. Dynamic simulation using MATLAB/Simulink, Python, or other environments allows students to explore process models, test controller tuning, and visualize transient responses without the safety risks and the challenges of operating full-scale equipment (Park et al., 2020; Zhang et al., 2013). Low-cost hardware platforms, such as Arduino-based Temperature Control Labs that developed by Dr. Hedengren at Brigham Young University (Park et al., 2020), extend learning to the physical domain, giving students direct experience implementing PID algorithms and confronting real-world issues like noise, nonlinearity, and actuator limits (Higginson and Brooks, 2022). Open-source tools such as Jupyter notebooks have also been integrated into many process control curricula, enabling interactive, self-paced learning and reducing barriers to software access (Bascuñana et al., 2023). Beyond individual tools, problem-based learning, cooperative projects, and remote laboratory systems have broadened access to authentic control scenarios, further bridging the gap between theory and industrial practice (Park et al., 2020; Rodríguez et al., 2018). Collectively, these methods reflect a sustained movement toward more engaging, applied, and technologically integrated instruction in process control to prepare graduates for the increasingly data-driven and model-rich industrial environment.

Recent work shows that artificial-intelligence tools are starting to support process-control teaching in practical ways. In chemical engineering, instructors have asked students to use a large-language-model (LLM) to help build simple dynamic models and then test them in the lab. students learned how to check AI outputs and use the tool ethically when deriving and validating tank-drain models before a hands-on session (Keith et al., 2025). Beyond single activities, Tsai and co-authors used ChatGPT to guide step-by-step problem modeling across core courses, which indicated that LLMs can scaffold coding, equation setup, and interpretation if teachers provide clear prompts and verification steps (Tsai et al., 2023). Broader education studies also indicate that adding ChatGPT to a flipped format can increase learning gains versus video-only preparation in undergraduate engineering courses, although careful structure and monitoring are important (Huesca et al., 2024). In control education outside the chemical engineering field, one study reported that ChatGPT was able to complete and pass the tasks in an undergraduate control course. However, the result also showed that such performance may come from surface-level understanding. Therefore, it

is important to provide clear guidance, such as using scoring rubrics, testing the code, and confirming understanding through oral questioning (Puthumanai et al., 2025). Surveys and primers aimed at chemical engineers provide practical guidance on when to use LLMs and how to verify outputs, emphasizing data security and accuracy (Dee and Faraji, 2024). Finally, reviews connecting modern AI methods to process control (e.g., reinforcement learning) give instructors context for introducing AI topics alongside PID/MPC, while noting stability, constraint handling, and safety as key teaching points (Faria et al., 2022).

Building on these recent developments in technology-enhanced and tutor-assisted process control education, we initiated a pilot project at Villanova University that applied large language models as virtual tutors to guide students in deriving and simulating classical process control models. The chemical engineering program at Villanova follows a traditional curriculum in which the process control course places strong emphasis on mathematical derivations and manual coding in MATLAB. Many students find it difficult to connect abstract equations with their physical meaning and often encounter challenges when debugging code. The shift to online teaching during the COVID-19 pandemic further highlighted the need for interactive and accessible learning materials. In recent years, Python has emerged as a widely used open-source alternative to commercial software, gaining popularity in engineering, data science, and related fields (Ozgur et al., 2021; Siva and Yamaganti, 2019). Web-based platforms such as Google Colab allow students to run Python code directly in a browser without software installation (Vallejo et al., 2022). Language-based computing assistants can also offer step-by-step explanations and generate example code for problem solving (Kingchang et al., 2023; Silva et al., 2024). Building on these developments, we designed a three-module pilot project that incorporated AI-assisted learning through Python in Google Colab into the process control course. The objectives were to: (1) help students translate physical principles into ordinary differential equations (ODEs) and simulate them; (2) clarify the use of Laplace transforms and transfer-function analysis; (3) provide guided practice in proportional–integral–derivative (PID) controller tuning; and (4) gather feedback on students' learning experiences.

2. Materials

2.1. Handouts and instructional materials

Three comprehensive handouts were prepared, each corresponding to a module (details summarized in Table 1). Module 1 (ODE simulation) introduced the formulation of mass and energy balances and numerical integration using Euler and Runge–Kutta schemes. The module provided students with an opportunity to review differential equations learned in the previous year, using AI assistance to connect chemical engineering principles, such as mass balances and chemical kinetics, to ordinary differential equation (ODE) systems, including liquid-level regulation in a water tank (Fig. 1 A), heated tanks (Fig. 1B), and enzyme kinetics (Fig. 1 C). Students asked ChatGPT to explain the physical meaning of terms in ODE equations, provide guidance on Python coding for ODE model simulation, and suggest Python code for numerical integration. Module 2 (Laplace and transfer functions) provided students step-by-step examples on performing Laplace and inverse Laplace transforms and deriving transfer functions from differential equations. The handout guided students in constructing closed-loop transfer functions, evaluating system response, and emphasizing the interpretation of gain, time constants, and poles. With the support of ChatGPT, students then developed Python programs to determine the right substrate concentration to produce the target cell contraction in a bioreactor (Fig. 1D). As for the closed-loop stability, students simulated a jacketed continuous-flow stirred-tank reactor (CSTR), as shown in Fig. 1E, and compared the responses of P controllers with several given proportional gains to examine how proportional gain affects the stability of the closed

Table 1
Overview of instructional materials.

Module	Handout focus	Key activities	AI prompts
1. ODE simulation	Mass and energy balances; Euler and Runge–Kutta integration; chemical kinetics; energy balance	Derive ODEs for tank-level control, heated tank and enzyme kinetics; implement numerical integration; plot time profiles	Ask ChatGPT to check ODEs, explain variables, provide Python code, compare results
2. Laplace and transfer functions	Laplace and inverse Laplace transforms; transfer-function derivation; closed-loop simulation; impact of proportional gains	Transform differential equations to Laplace domain; derive transfer functions; estimate parameters from data; simulate the closed-loop transfer function with P controllers of different proportional gains	Ask ChatGPT to perform Laplace transforms, derive transfer functions, interpret poles and zeros, suggest controller gains
3. PID tuning	PID algorithm implementation; tuning rules (ITAE-rule); advanced controller features	Write general PID code; tune parameters; compare controller performance	Ask ChatGPT to generate and debug PID code, propose tuning parameters, explain effect of proportional, integral and derivative terms

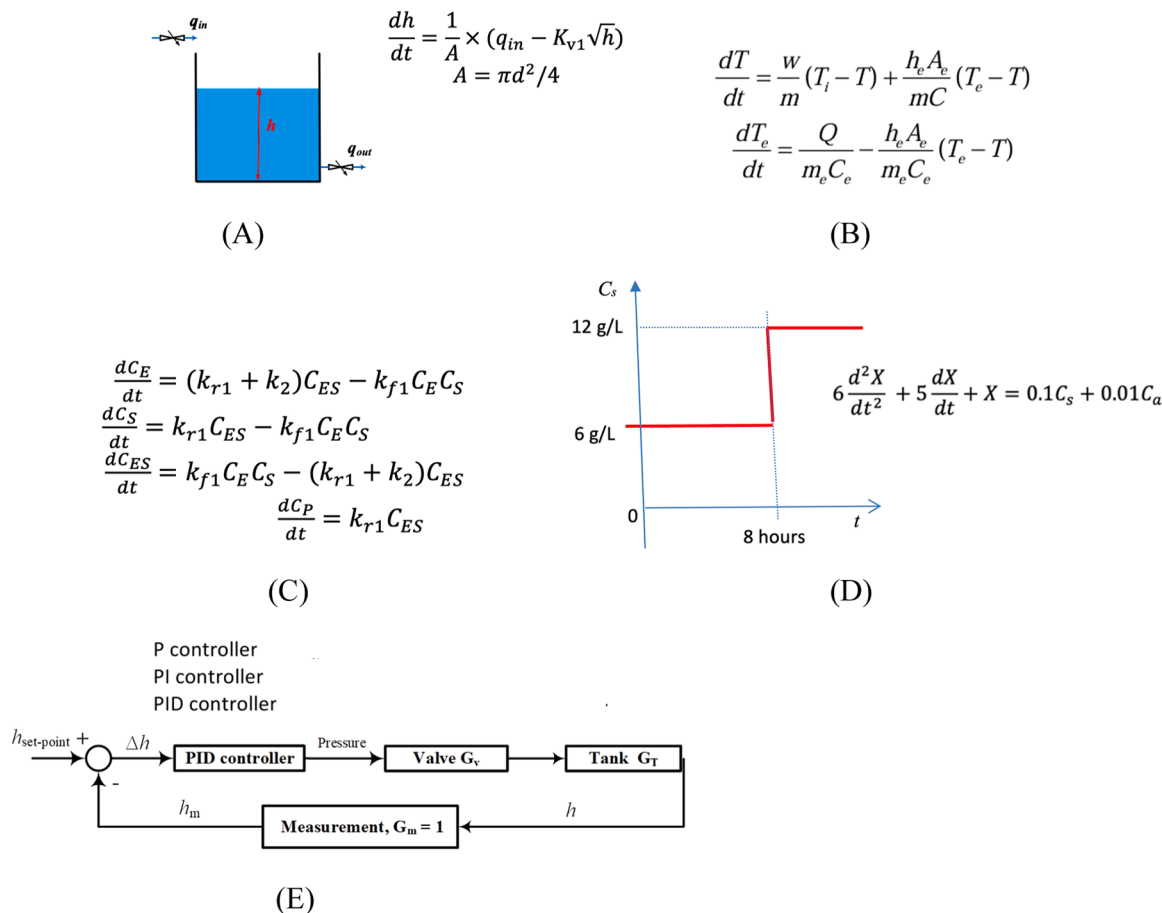


Fig. 1. (A) the ODE model for a water tank in which q_{in} is inlet flowrate, m^3/min , K_{v1} is the valve coefficient, d is the diameter of the tank; (B) the ODE model for an electrically-heated tank in which T is the temperature of the process fluid in the stirred tank, T_e is the temperature of the heating element, t is time, T_i is the inlet fluid temperature, and Q is the electrical heating rate supplied to the heater, m and C represent the mass flow rate and heat capacity of the fluid, m_e and C_e denote the mass and heat capacity of the heating element, h_e is the heat transfer coefficient between the heater and the fluid, and A_e is the surface area for heat exchange; (C) the ODE model for an enzymatic process that converts substrate S to the product P catalyzed by the enzyme E , with k_{f1} , k_{r1} , and k_2 are reaction constants; (D) the ODE model for a bioreactor to predict the concentration of the cell concentration (X) for a step change in the substrate concentration (C_s) and a constant amino acid concentration (C_a), through Laplace transform, transfer function, inverse Laplace transforms; (E) An exercise for students to compare the stability of P controllers with different proportional gains and then to tune and compare P, PI, and PID controllers for a given water tank (G_T) and a given control valve (G_v).

loop to control the liquid level in a tank. In all these exercises, students were encouraged to develop their own Python programs based on the examples provided in the handout and to correct or refine their code with the support of ChatGPT. Module 3 (PID tuning) focused on PID controller design. Different from Module 2 in which the proportional gains were given, students were required to develop their own PID controllers using the knowledge they learned from class. They then use ChatGPT to validate their controller design using the ITAE tuning rules. They implemented general PID algorithms in Python (with PI and PID added in Fig. 1E), explored ITAE tuning rules by visualizing the output

profile in Python. They then refined or improved their programs with the support of ChatGPT. As part of the authentic activity, each student was required to prompt ChatGPT to modify the problem statement and generate a real-world application challenge, solve the new problem, and record their solutions. The real-world challenges created by ChatGPT differed for each student. For each module, a pre-class video introduced the theoretical background and demonstrated the use of ChatGPT. Students completed written reports summarizing their derivations, simulation results and reflections on AI assistance.

2.2. Teaching approach and implementation

This pilot project was part of CHE4202 Chemical Process Control at Villanova University in Spring 2025, a senior-level course. That semester, 59 students registered, completed all three learning modules, and submitted assignments; 47 responded to the anonymous survey. Students came with limited programming experience, and some of them were learning Python for the first time.

Each module was assigned after the relevant theory had been introduced in class and the corresponding homework completed. The modules were designed to help students review previously learned material and connect it to real-world applications. The course followed a quasi-flipped-classroom design: before each module session, students received handouts and short lecture videos prepared by the instructor. These videos reviewed related theory and demonstrated example Python programs in Google Colab. Students were encouraged to pause the videos and try the examples themselves. During class, the instructor held a brief (~10 min) Q&A session to address questions and highlight common mistakes or issues identified during office hours. The instructor also demonstrated how to provide effective ChatGPT prompts to help write equations, explain concepts, generate code, or fix programming errors.

The AI training approach evolved across the three modules. In Module 1, students were given example prompts and relied heavily on AI to learn ODE terminology, interpret equations, and develop simulation code from scratch. Informal feedback during conversations revealed that many students struggled to determine whether AI-generated Python programs were correct. To address this, Modules 2 and 3 incorporated a more guided structure: the instructor provided example Python programs, and assignments required students to adapt and modify these programs for new problems. This approach allowed students to focus on interpreting and applying code while gradually building their ability to

verify correctness.

After each module, students submitted written reports and Python scripts. Grading considered the correctness of equations, quality of code, clarity of plots, and reflections on their use of AI to solve real-world problems generated by ChatGPT. Timely feedback enabled students to improve their work. The instructor emphasized that ChatGPT should serve as a virtual tutor rather than an unquestioned authority, and students were expected to check programs using AI or critically review AI-generated outputs before submission.

2.3. Survey instrument and data collection

To evaluate the learning impact and overall experience of AI-assisted modules, we designed a 15-question anonymous survey. Questions 1–8 assessed the extent to which the project improved students' understanding of ODE modeling, numerical methods, Laplace transforms, controller design and related topics. Questions 9–13 evaluated students' perceptions of engagement, clarity of instructions, problem-solving skills and applicability to real-world problems. Questions 14–15 were open-ended questions on what students liked most and what they found challenging or would improve. Each closed-ended item was rated on a five-point Likert scale (1 = strongly disagree, 5 = strongly agree). The survey instrument is reproduced in Table 2. Survey responses were collected in class and converted to a CSV file for statistical analysis.

3. Results

3.1. Instructor evaluation of student performance

The instructor carefully evaluated students' reports on the solutions, the Python programs, and their solutions to the real-world problem created by ChatGPT. Overall, students demonstrated strong technical

Table 2
Survey questionnaires.

Section 1: Learning Impact 1. The AI-based project helped me better understand how to model chemical processes. ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 2. The project improved my ability to solve ODEs using numerical and analytical methods. ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 3. The project made the concept and application of the Laplace Transform clearer. ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 4. I gained a better understanding of Inverse Laplace Transform through the simulation exercises. ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 5. The project showed me how Laplace analysis applies to chemical processes. ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 6. The second project improved my understanding of PID controller design. ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 7. I can now better interpret how changing controller parameters (K_c, τ_I, τ_D) affects system performance (e.g., offset, oscillation). ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 8. The hands-on coding aspect helped reinforce theoretical concepts taught in class. ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 Section 2: Overall Learning Experience 9. The AI-based projects were engaging and motivated me to learn. ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 10. The use of Python-based simulations enhanced my problem-solving skills. ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 11. I found the instructions and handouts clear and helpful for completing the projects. ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 12. These projects helped me see how AI and programming can be applied in chemical engineering. ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 13. I would recommend these AI-based projects to students in future classes. ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 Section 3: Open-Ended Questions 14. What did you like most about the AI-based projects? 15. What did you dislike or find challenging about the AI-based projects? Suggestions for improvement?

skills introduced in the Chemical Process Control class by implementing the modules with the support of AI. Most were able to complete the required tasks successfully and apply the concepts to solve problems within the deadline. Only about five students (out of 59 students) earned low scores, in the range of average 51–78 in the three modules. Among them, two students did not finish Module 1. This might be due to the challenges with either the underlying concepts or the application of AI tools. But these two students made significant improvement afterward.

3.2. Students' self-evaluation from the survey

Forty-seven of these responded to the anonymous survey. The response rates summarized in Table 3 shows the mean $\pm$ standard deviation for each survey item. Survey results indicated that students found the AI-based modules beneficial across multiple learning objectives, though the degree of impact varied by topic. Moderate gains were reported in understanding ODE modeling and solving ODEs using numerical and analytical methods (both 3.57), while improvement of understanding on the Laplace Transform was rated slightly lower (3.36) and the Inverse Laplace Transform received the lowest score (3.28). This suggests these areas may require additional instructional support. Students perceived strong real-world relevance in applying Laplace analysis to chemical processes (3.79) and reported improved understanding of PID controller design (3.66), with particularly high ratings for interpreting the effect of changing controller parameters (3.85). The hands-on coding component (3.70) was valued for reinforcing theory, and the AI-based projects were seen as both engaging and motivating (3.85). Python-based simulations were credited with enhancing problem-solving skills (3.81), while instructions and handouts were rated as clear and helpful (3.94). The highest score (4.36) reflected the projects' effectiveness in demonstrating how AI and programming can be applied to chemical engineering, and students expressed strong willingness to recommend the projects for future classes (4.23). Overall, students reported moderate to high agreement with statements indicating improved understanding and engagement. These results suggest that students recognized the value of AI assistance in helping them apply concepts to practical scenarios but still relied on instructor guidance, particularly for challenging concepts like Laplace and inverse Laplace transform.

Table 3
Summary statistics for survey items.

Question	Mean $\pm$ SD
Q1. The AI-based project helped me better understand ODE modeling.	3.57 $\pm$ 0.9
Q2. The project improved my ability to solve ODEs using numerical and analytical methods.	3.57 $\pm$ 0.85
Q3. The project made the concept and application of the Laplace Transform clearer.	3.36 $\pm$ 1.09
Q4. I gained a better understanding of Inverse Laplace Transform through the simulation exercises	3.28 $\pm$ 1.04
Q5. The project showed me how Laplace analysis applies to chemical processes.	3.79 $\pm$ 0.95
Q6. The second project improved my understanding of PID controller design	3.66 $\pm$ 1.13
Q7. I can now better interpret how changing controller parameters affects system performance.	3.85 $\pm$ 0.96
Q8. The hands-on coding aspect helped reinforce theoretical concepts.	3.7 $\pm$ 1.1
Q9. The AI-based projects were engaging and motivated me to learn.	3.85 $\pm$ 0.96
Q10. The use of Python-based simulations enhanced my problem-solving skills.	3.81 $\pm$ 1.04
Q11. I found the instructions and handouts clear and helpful.	3.94 $\pm$ 0.94
Q12. These projects helped me see how AI and programming can be applied in chemical engineering	4.36 $\pm$ 0.67
Q13. I would recommend these AI-based projects to students in future classes.	4.23 $\pm$ 0.87

3.3. Students' feedback via open-ended questions

Student feedback highlighted several recurring strengths of the AI-based modules (Table 4). Many students valued the opportunity to develop coding and Python skills (13 mentions) and appreciated the exposure to AI tools such as ChatGPT and Gemini (10 mentions). The real-world application of concepts (8 mentions) and the ability to reinforce understanding of chemical engineering principles were also frequently cited. Several students noted the novelty and engaging nature of the assignments (7 mentions) as well as their efficiency and ease of completion (4 mentions). While only a few students mentioned the usefulness of videos and instructional clarity (2 mentions), those who did found them highly beneficial.

Open-ended comments were further grouped into positive aspects and challenges (Table 5). Positive comments are consistent with those listed in Table 4. For example, the project provided something different from typical homework, increased engagement, and demonstrated practical applications. Students appreciated being able to ask ChatGPT for quick explanations and code, which allowed them to explore new problems that would have been difficult alone. Real-world examples made learning more interesting and relevant. Some students noted that AI assistance reduced frustration when debugging. On the other hand, challenges included unclear instructions, insufficient coding background, and occasional errors in AI-generated code. Several students wanted more explanation in class and more practice problems. A number of respondents found it difficult to judge whether ChatGPT's outputs were correct and sought more guidance on verification.

4. Discussion

4.1. Positive feedback on AI-assisted learning in chemical process control

The results of this pilot project demonstrate that AI-assisted modules can be effectively integrated into a senior-level chemical process control course to support both technical skill development and student engagement. Instructor evaluations showed that most students were able to complete the required tasks successfully, apply process control concepts to real-world problems, and produce functional Python simulations. Survey results (Table 3) revealed moderate to high gains across multiple learning objectives, with the highest ratings for understanding how AI and programming can be applied to chemical engineering (4.36) and willingness to recommend the modules to future cohorts (4.23). Open-ended feedback (Tables 4 and 5) reinforced these findings, showing that students particularly valued the development of coding skills, exposure to AI tools, and the relevance of assignments to real-world applications.

4.2. Challenges in using AI for teaching chemical process control

Implementing AI in the classroom presented certain major challenges, particularly in the areas of prompt preparation and code verification. In Module 1, students relied heavily on AI for learning ODE concepts and generating Python simulations. Although example prompts were provided, conversations with students revealed

Table 4
Frequency of themes identified in student feedback on AI-based modules.

Theme	Count
Coding and Python skills	13
AI exposure and tools	10
Real-world application	8
Conceptual understanding	8
Novelty and engagement	7
Efficiency and ease	4
Videos and instructional clarity	2

Table 5

Selected positive and negative comments from the open-ended survey questions.

Category	Sample comments
Positive	"It was something different from our other homework." "The examples and videos were very helpful; I could follow step by step." "Using ChatGPT helped me debug quickly." "Real-world problems made the learning more interesting."
Negative	"More explanation of the code would help my understanding." "Sometimes AI gave wrong code; I needed more guidance to check." "I was slow because my Python background was weak." "It was hard to decide if the AI answer is correct." "I needed more practice problems to feel confident."

difficulties in determining whether AI-generated code was correct. This aligns with survey and comment data indicating uncertainty about verification and a need for more explicit guidance. To address this, Modules 2 and 3 introduced a more structured approach by providing example Python programs and requiring students to adapt them for new problems. This scaffolded method reduced the risk of propagating errors from unverified AI outputs and improved students' confidence in modifying and testing code. However, the persistence of comments about AI-generated inaccuracies underscores the importance of teaching students not only how to query AI effectively but also how to critically assess its responses. While students were suggested by the instructor to use other AI tools, such as Gemini, to cross-validate the programs generated by ChatGPT, inconsistency issues still persisted.

4.3. Limitation and future work

Despite overall positive outcomes, certain limitations were evident. Lower ratings for improved understanding the Laplace Transform (3.36) and Inverse Laplace Transform (3.28) suggest that these mathematically intensive topics require more targeted instructional support. Variability in students' Python backgrounds also affected performance, with some participants reporting that a lack of prior coding experience slowed their progress. Additionally, AI integration was limited to a single course and semester, which restricts conclusions about long-term learning gains or transfer of skills to other contexts. Future iterations of the project will focus on three key improvements. First, developing structured prompt-writing exercises and verification checklists will help students use AI more effectively and critically. Second, incorporating additional in-class coding practice and peer review activities could provide real-time opportunities to identify and correct AI-generated errors. Third, expanding the approach across multiple courses and tracking skill retention over time will provide stronger evidence of the long-term benefits and limitations of AI-assisted learning. Broader adoption could also explore integrating AI into laboratory-based courses to enable students to connect computational modeling with hands-on process control experiments.

5. Conclusion

This pilot study demonstrates that AI-assisted modules can be successfully integrated into a senior-level chemical process control course to enhance both technical competence and student engagement. By combining pre-class instructional materials, AI assistance, and targeted coding activities, students were able to apply ODE modeling, Laplace transforms, and PID controller design concepts to real-world problems with the support of tools such as ChatGPT. Survey data and open-ended feedback revealed that students valued the opportunity to develop coding skills, explore AI applications, and engage with assignments that connected theory to practice. The project also highlighted important instructional considerations for AI-based learning. Students initially struggled with prompt formulation and verifying AI-generated code. This implied the need for explicit training in both effective AI use and evaluation of outputs. A progressive instructional design, i.e., shifting

from AI dependence in Module 1 to scaffolded code modification in Modules 2 and 3, helped address these challenges and improved students' ability to adapt and validate computational solutions. While the results are promising, limitations such as varying levels of coding proficiency and lower perceived gains in mathematically intensive topics suggest that further refinements are needed. Future work will focus on strengthening verification skills, expanding in-class practice opportunities, and extending AI integration to other courses and contexts. Overall, this study provides a replicable framework for incorporating generative AI into engineering education, offering a pathway to prepare students for a future where computational tools and human expertise work hand in hand.

CRediT authorship contribution statement

Zuyi (Jacky) Huang: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization.

Declaration of Competing Interest

There is no conflict of interest in this project.

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